

Case 02: Finite-Duration Firebrand Generation

1 Verification purpose

Case ID: CASE02_FGR

This case verifies finite-duration firebrand generation when the source lifetime is not commensurate with the solver timestep. It compares timesteps shorter and longer than the residence time while holding the source, grid, wind, and transport distribution fixed.

The test exposes whole-step over-generation, truncation of the last active interval, and timestep-dependent deposition. Success requires both variants to recover the same integrated source count and downwind profile within tolerance.

2 Mathematical formulation and numerical implementation

Let a burning computational cell generate firebrands at the constant rate GR from ignition time t_{ign} until $t_{\text{ign}} + t_{\text{res}}$. The number generated during an ELMFIRE step $[t, t + \Delta t]$ must use the overlap between that step and the active source interval:

$$\Delta t_{\text{active}} = \max(0, \min(t + \Delta t, t_{\text{ign}} + t_{\text{res}}) - \max(t, t_{\text{ign}})) ,$$
$$\Delta N = \text{GR } \Delta t_{\text{active}} .$$

This formulation handles the first and final partial steps and also handles a single timestep that spans the entire residence time. Summing over all steps must give

$$N_{\text{total}} = \text{GR } t_{\text{res}} .$$

The case sets $\text{GR} = 10.0$ pcs/s and $t_{\text{res}} = 10.0$ s, so the expected total is 100.0 firebrands.

2.1 Implementation path and verification rationale

The documented finite-duration source formulation identifies a timestep-integration error when the solver step is comparable to or longer than the fireline residence time [1]. In current ELMFIRE, READ_SPOTTING selects the PER-AREA source, SPOTTING computes the active-cell emission, and the source timing state maintained by CALC_SPOTTING_DURATION bounds that emission. The accumulated Eulerian field is advanced by EMBER_TRAJECTORY_EULERIAN. Comparing a fine step with a step longer than the complete source lifetime directly tests whether partial-step overlap, rather than the nominal solver step, controls the emitted amount. The total-count metric detects temporal over- or under-integration; the profile additionally detects correct totals deposited in wrong cells. Surface spread, ignition, consumption, and crosswind dispersion are disabled, so those processes cannot compensate for a source-timing error. The test verifies numerical accounting and the configured transport kernel, not the physical realism of the prescribed lognormal distribution.

3 Derivation of expected behavior

Generated firebrands are distributed with a lognormal distance model using $\mu = 2.18$ and $\sigma = 1.23$. The distribution is truncated at its 99th percentile. ELMFIRE rounds that distance to an integer number

K of cells and normalizes the retained probability:

$$P_i = \frac{F(i\Delta x) - F((i-1)\Delta x)}{F(K\Delta x) - F(10^{-6})}, \quad i = 1, \dots, K,$$

where F is the lognormal cumulative distribution. The analytical count in cell i is $N_i = 100P_i$. The post-processor reproduces this cell-integrated reference and compares it directly with the final `ember_flux` raster.

The two time integrations can be reproduced explicitly. For $\Delta t = 0.13$ s, 76 complete steps cover $76(0.13) = 9.88$ s and the final overlap is $10 - 9.88 = 0.12$ s, so

$$N_{\text{fine}} = 10[76(0.13) + 0.12] = 100.$$

For the $\Delta t = 12.7$ s member, the first step spans the complete source interval and

$$\Delta t_{\text{active}} = \min(12.7, 10) - \max(0, 0) = 10 \text{ s}, \quad N_{\text{coarse}} = 10(10) = 100.$$

For the spatial reference, $z_{0.99} = 2.3263479$ gives

$$x_{0.99} = \exp[2.18 + 1.23(2.3263479)] = 154.6876 \text{ m}, \quad K = \text{NINT}(154.6876/10) = 15.$$

The retained CDF mass is $F(150) - F(10^{-6}) = 0.9893135$. Substitution in the cell formula gives $N_1 = 54.5524$, $N_2 = 20.8937$, $N_3 = 9.4216$ particles, with all 15 bins summing to 100 after normalization. Hence the exact metric references are $E_N = 0$ and $E_{L_1} = 0$; the numerical tests substitute $E_N \leq 0.005$ and $E_{L_1} \leq 0.01$ for both timesteps.

4 Verification metrics and acceptance criteria

The integrated reference count is $N_{\text{ref}} = G_R t_{\text{res}} = 10 \times 10 = 100$ particles. The analytical spatial reference integrates the configured lognormal landing distribution over each downwind cell, truncates it at the 99th percentile, and renormalizes the cell probabilities to N_{ref} . The ignition row and physical columns are read from each manifest; all halo cells and nodata values are excluded.

4.1 Justification of metric quantification

Both limits have a direct particle-count interpretation. Since $N_{\text{ref}} = 100$, the integrated-error bound $E_N \leq 0.005$ permits an absolute imbalance of only 0.5 particle. This is large enough for raster roundoff and percentile-edge binning, but too small to hide a one-particle source-duration error. The normalized L_1 bound of 0.01 permits at most one particle's worth of total spatial redistribution after the analytical lognormal probabilities have been integrated over the same cells. Unlike a signed error, this norm cannot conceal compensating upstream and downstream errors. Applying both conditions to the short- and long-timestep variants separates conservation of the total from correct overlap-limited placement. The exact variant count is required because the intended assertion concerns both sides of the residence-time boundary.

4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	One final <code>ember_flux</code> raster with current manifest geometry for each timestep.	2 of 2 variants usable
Integrated-count error	$E_N = \sum_{ij} N_{ij} - N_{\text{ref}} /N_{\text{ref}}$, using the full valid output raster.	$E_N \leq 0.005$ for each timestep
Profile normalized L_1 error	$E_{L_1} = \sum_k N_k - N_{k,\text{ref}} /N_{\text{ref}}$ on matched physical downwind cells.	$E_{L_1} \leq 0.01$ for each timestep
Overall decision	Conjunction of output completeness and both errors for both timestep variants.	PASS only if every component passes

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for `dt0p13`. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

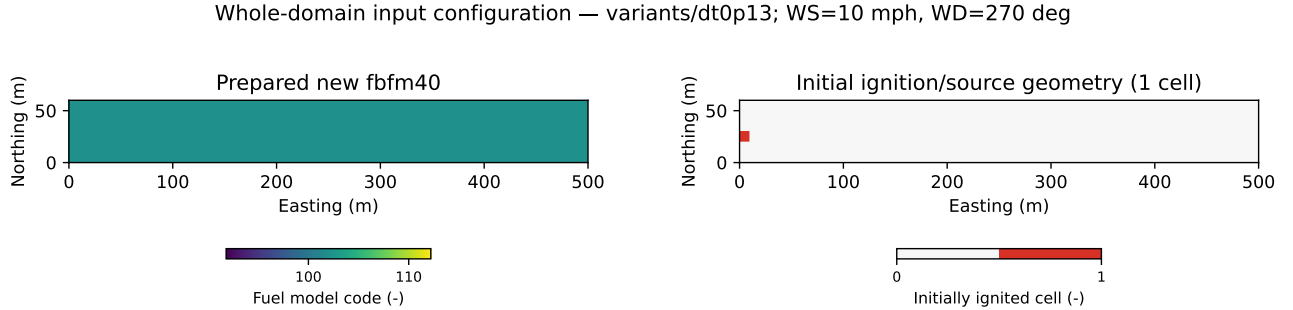


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The preprocessor creates two variants with $\Delta t = 0.13$ s and $\Delta t = 12.7$ s. Because timestep variation is the quantity under test, both fractional values are retained. The requested 120-s stop is rounded upward to an exact multiple: 924 steps give 120.12 s for the 0.13-s member, and 10 steps give 127.0 s for the 12.7-s member. The dump interval is set to the same aligned stop, while the verified 10-s source duration is unchanged. Both use a 10 m grid and the same 500 m by 60 m physical domain. This gives 50 by 6 usable cells and a 54 by 10 raster after two numerical buffer cells are added outside all four sides. A single ignition is placed in the first usable column and the middle usable row.

The source is isolated with `NO_SURFACE_FIRE=.TRUE.`, so it does not create additional burning cells. The current PER-AREA generation model calculates

$$\text{GR}_{\text{pixel}} = \text{EMBER_GR} \Delta x \Delta y.$$

Using `EMBER_GR=0.1` pcs/s/m² and a 100 m² cell gives the required 10 pcs/s source. The fixed source duration is selected with `USE_PHYSICAL_SPOTTING_DURATION=.FALSE.` and `TAU_EMBERGEN=10` s. Firebrand ignition, consumption, and crosswind dispersion are disabled so that they cannot change the accumulated count.

Each variant must satisfy both conditions:

- total-count relative error no greater than 0.5%;
- profile L_1 relative error no greater than 1%.

The overall case passes only if outputs exist and both timestep variants pass. The two-cell buffers are excluded from profile extraction, while the total count is checked over the complete raster to detect misplaced firebrands.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from dt0p13. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2} .

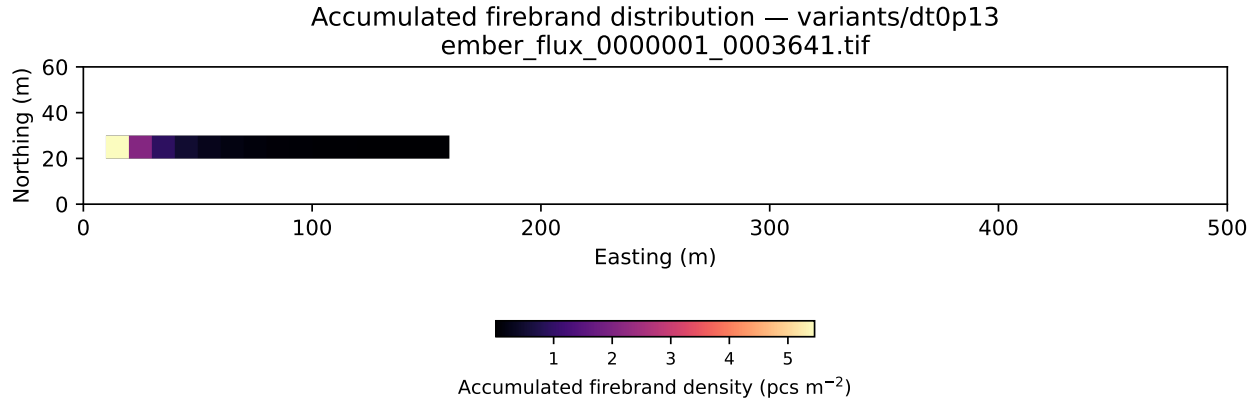


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Overall status: **pass**. Figure 3 first compares each actual ELMFIRE profile with the cell-integrated analytical distribution. Agreement for the 12.7 s variant is the critical result because that timestep exceeds the complete 10 s source duration.

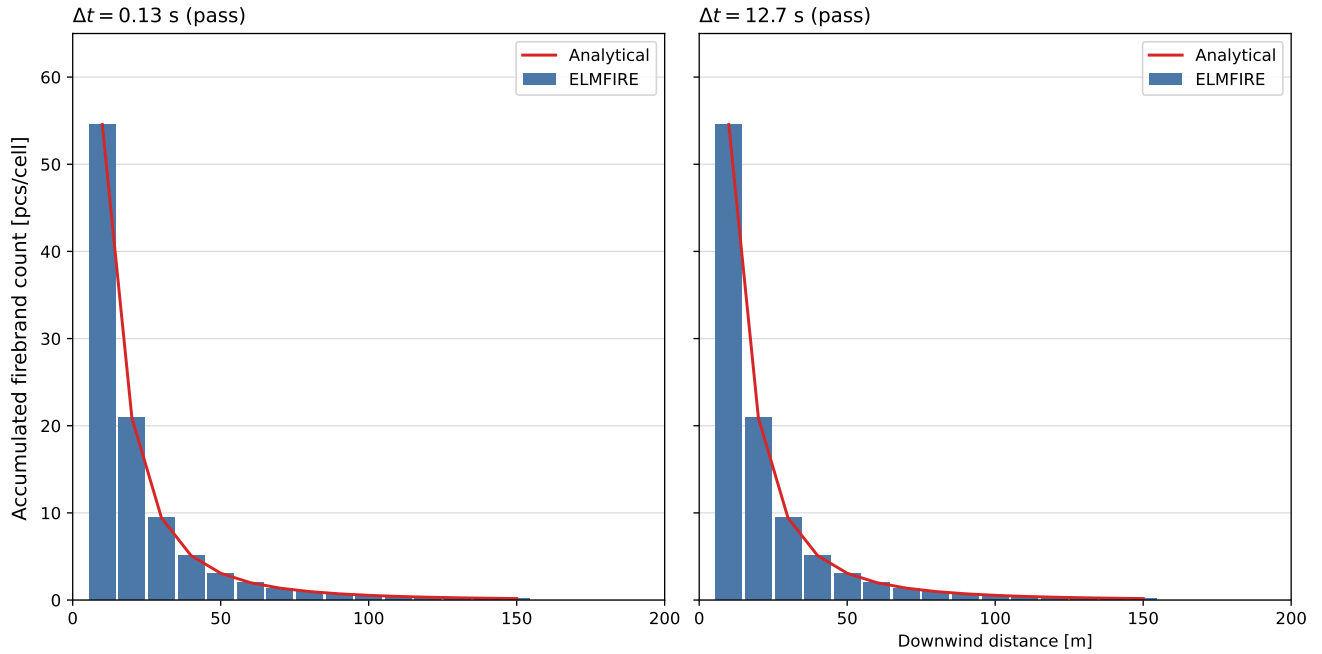


Figure 3: Accumulated firebrands per downwind cell for the 0.13 and 12.7 s timestep variants. Blue bars are ELMFIRE output and the red curve is the cell-integrated, truncated-lognormal reference for 100 generated firebrands.

7 Calculated metrics and verification decision

Timestep	Observed total	Total relative error	Profile L_1 error
0.13 s	99.9999	5.198e-07	8.965e-07
12.7 s	100	2.399e-08	3.449e-07

Table 2: Firebrand-count and spatial-distribution errors for the two timestep variants.

The overall decision is **pass** and is taken directly from postprocessing.

8 Technical interpretation

Agreement of both totals and profiles across a timestep longer than the complete source duration demonstrates correct partial-step accounting; a discrepancy isolates temporal source integration or deposition indexing.

9 Scope and limitations

The generated variant namelists, raster dimensions, buffer width, source location, rates, duration, and analytical parameters are recorded in `../variants/manifest.json`. Per-variant raster filenames and numerical errors are recorded in `../outputs/metrics.json`.

References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.